

Vignesh Mohanavelu

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EDUCATION

Arizona State University, MS in Computer Science

Aug. 2024 – May 2026 | AZ, USA

Relevant Courses: *Algorithms, NLP, Cloud Computing, Advanced OS*

GPA: 4.0/4

National Institute of Technology, B.Tech in EE

Jul. 2018 – May 2022 | Trichy, India

Relevant Courses: *Data Structures, Operating Systems, DBMS*

(First Class)

PROFESSIONAL EXPERIENCE

Software Engineer - Full Stack

Jul 2022 – Aug 2024

Standard Chartered Global Business Services

Chennai, India

- Co-owned development of an enterprise trade platform in an **Agile** environment using **React, Node.js, Spring Boot, Oracle DB**, handling **2M+** daily transactions.
- Microservice Architecture**: Orchestrated migration from a legacy monolithic system to event-driven microservices using Apache **Kafka**, reducing API response time by 45% and achieving sub-100ms message latency.
- Distributed Caching**: Designed a distributed caching service that serves as the primary market data for all trades across the system, increasing system throughput by 25% and reducing average query time by 35%.
- Security & Code Quality**: Implemented **OAuth2** based authentication and remediated 150+ vulnerabilities (XSS, SQL injection) through automated security scanning with SonarQube; added unit/integration tests with **Jest**, improving code quality by 60%.

Research Software Developer

Mar 2025 – Present

Biodesign Institute, Arizona State University

Tempe, AZ

- Architected an admin management platform using **Next.js, Express.js** and **PostgreSQL** with JWT based authentication to consolidate SharePoint workflows, reducing manual processing by 1.5x and approval turnaround by 60%.
- Integrated an text-to-SQL agent on Google ADK with a schema-aware MCP server, using constrained decoding to guarantee valid SQL and reach near 100% accuracy on internal queries.
- Modernized a cron-based polling pipeline into a real-time, event-driven **MQTT** ingestion system streaming telemetry from 350+ lab freezers through a pub/sub broker, cutting costs by over **90%**.

Software Analyst Intern

May 2021 – Jul 2021

Hexaware

Bangalore, India

- Built **REST APIs** using **FastAPI** to integrate supply-chain microservices, parallelizing independent service calls with goroutines to raise throughput by **20%**.
- Introduced and maintained **Swagger/OpenAPI** documentation to lower API integration errors by 15%.

TECHNICAL SKILLS

Languages : C/C++, JavaScript/TypeScript, Java, Python, HTML/CSS

Web Frameworks : React, Angular, Node.js, FastAPI, Express, Spring Boot, MongoDB, Kafka, PostgreSQL

AI/ML : Pandas, Scikit-learn, TensorFlow, PyTorch, LangChain, RAG, Vector databases

Cloud & DevOps : AWS (EC2, S3, Lambda, EKS, Bedrock), Azure, Docker, Jenkins, Git, CI/CD, Agile

PROJECTS

Multi-Agent Legal Discrepancy Detection 🔗 | *LangGraph, Pydantic, RAGAS, Qdrant*

Jan 2025

- Built a multi-agent **LangGraph** workflow using **RAG** and **Qdrant** to detect discrepancies in legal contracts.
- Developed orchestration, compliance, and clause-rewriting agents using **GPT-4o** and **Gemini**, with structured **Pydantic** outputs for confidence-based review.
- Created a **RAGAS** evaluation pipeline to measure retrieval quality on labeled contract datasets.

TCR-Epitope Binding Affinity Prediction 🔗 | *Python, LLM, BERT, catElMo*

Dec 2025

- Trained BERT and catElMo sequence embeddings with neural classifiers on **300K+** TCR-epitope pairs, reaching strong AUC and F1 under TCR-split and epitope-split protocols on unseen sequences.

DineSphere Chatbot | *Python, Strands, AWS EKS, Bedrock, OpenSearch, DynamoDB, SES*

2026

- Built a cloud native AI restaurant recommender on **EKS** that turns a natural conversation into personalized dining suggestions, backed by a hybrid **RAG** retriever (**OpenSearch** BM25 + vector search with reranking).
- Drove the dialogue with a stateful agent (tool-calling, conditional routing, **DynamoDB** session state) and offloaded recommendation emails to an async **EventBridge**, Step Functions, and SES pipeline.